1 Orchestrate or Upgrade? The Cost-Quality Economics of Multi-Agent Pipelines versus Frontier Models — When Lower Tier Teams Deliver 98% Quality at 23% Cost

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Version: v3 (2026-08-09) — Business of AI edition: business-oriented framing with full academic rigor preserved.

1.1 Abstract

1.2 Abstract

Organizations deploying large language models (LLMs) for knowledge work face a strategic choice: invest in higher-tier models or invest in multi-agent pipeline structures that orchestrate cheaper models. We test this resource-allocation tradeoff with a pre-registered 2 (model tier) × 2 (pipeline structure) × 2 (persona) factorial experiment across five LLM families, 38 business-strategy problems, and 1,900 generation–judgment runs evaluated by a blinded three-judge panel. Our findings yield a decision framework for AI investment. First, team structure delivers higher marginal returns at the cheap tier than at the frontier tier ($d = -0.227$, $p = 0.015$): a cheap model with a three-step pipeline approaches frontier quality at 23% of the cost (T-EGR = 6.15). Second, personas degrade output quality on average ($d = -0.691$), representing a failed investment in stylistic differentiation. Third, a low-cost refinement pass recovers 60–64% of the frontier gap but never achieves paired superiority — a bounded-improvement result that constrains the substitution thesis. Qualitative coding (= 0.824) reveals that orchestration buys coverage (more structure, more length) while the frontier wins on density (compressed insight, zero boilerplate). The Pareto frontier is stable across price assumptions, yielding actionable contingency rules for budget-constrained firms. We contribute the T-EGR efficiency metric, a mechanistic account of why structure substitutes for tier only partially, and a practitioner decision framework that maps budget tier and task type to optimal pipeline architecture.

Keywords: LLM Orchestration, Multi-Agent Pipelines, Cost-Performance Analysis, LLM-as-a-Judge, T-EGR, Pre-registration.

1.3 List of Abbreviations

2 1 Introduction

The past two years have seen a rapid industrial shift from single-model inference to the orchestration of multiple language-model calls: pipelines chain builder, critic, and refiner agents into structured workflows; agent teams assign social roles to each step; panels of judges aggregate verdicts across models. The implicit claim is economic — that *structure can substitute for model tier*. A practitioner who cannot afford a frontier model, the argument runs, can reach comparable output quality by composing cheaper models into an orchestrated system. If true, the claim reshapes how organizations price, budget, and architect generative-AI products. If false, orchestration is at best a marginal convenience and at worst a cost multiplier that never reaches frontier quality.

Prior empirical work on generative AI is encouraging but silent on this specific claim. Controlled productivity studies show that a single generative assistant raises output quality and reduces task time (Noy & Zhang, 2023; Brynjolfsson, Li, & Raymond, 2023), yet they treat the model as a given, single-agent capability. A field experiment with consultants documented a "jagged technological frontier": large gains inside the model's capability boundary, large losses outside it (Dell'Acqua et al., 2023). Neither study varies the structure around the model. Multi-agent and ensemble work shows that sampling, debate, and tree search can improve accuracy

Abbr	Definition
BF	Bayes factor in favor of H over H
C-S-N	cheap solo, no persona
C-S-P	cheap solo, persona
C-T-N	cheap team, no persona
C-T-P	cheap team, persona
F-S-N	frontier solo, no persona
F-S-P	frontier solo, persona
F-T-N	frontier team, no persona
F-T-P	frontier team, persona
EGR	Efficiency Gain Ratio
FDR	false discovery rate
FLOPs	floating-point operations
FSN	frontier solo, no persona (baseline)
ICER	incremental cost-effectiveness ratio
IRR	inter-rater reliability
JZS	Jeffreys–Zellner–Siow (Bayes factor prior)
LLM	large language model
LMM	linear mixed model
LOO	leave-one-out
OBF	O'Brien–Fleming (spending function)
SESOI	smallest effect size of interest
T-EGR	Tiered Efficiency Gain Ratio
TOST	two one-sided tests (equivalence)

(Wang et al., 2023; Liang et al., 2024; Yao et al., 2023), but the gains are reported without cost-matched comparison to a stronger single model. A nascent theory of test-time compute predicts that, for fixed problem difficulty, more inference compute on a smaller model can outperform less inference compute on a larger one (Snell et al., 2024), but its predictions are in units of FLOPs, not dollars, and only on verifiable tasks.

Three gaps follow. First, no factorial, cost-matched comparison of orchestrated cheap models versus frontier solo exists on open-ended tasks. Second, no mechanistic account explains *why* orchestration succeeds or fails — whether it adds structural coverage or compressed insight. Third, no economic metric is robust to the price-assumption choices that inevitably shape per-run costing.

This paper addresses all three. We run a pre-registered 2 (tier) × 2 (structure) × 2 (persona) factorial experiment over five model families, 38 domain-stratified business problems, and 1,900 generation–judgment runs, measured by a blinded three-judge panel from families disjoint from every generator. Cost is stored as tokens and re-priced through a frozen price vector under three blended input/output ratios, so every reported dollar is reproducible and re-priceable. A second study applies a cheap, add-only refinement pass to the orchestrated outputs and codes the qualitative differences between original, refined, and frontier answers, testing the mechanistic prediction that orchestration buys coverage while the frontier wins on density.

Five contributions follow:

1. First cost-matched pre-registered factorial on orchestration economics. Negative superiority results are reported as primary findings, not buried as null results.

2. **The T-EGR metric** (novelty ratio ÷ cost ratio) and a price-assumption-robust Pareto frontier.

3. **A mechanistic account** from a mixed-methods Study 2 (pooled = 0.824): orchestration buys coverage, the frontier wins on density.

4. **The refinement boundary**: a $0.002 refinement pass reaches parity with the weakest frontier but never superiority.

5. **Practitioner contingency rules** mapping cell to use-case to expected T-EGR.

The paper proceeds as follows. Section 2 reviews related literature. Section 3 describes the pre-registered design, apparatus, and analysis, including an AI-assisted research disclosure (§3.9). Section 4 presents the results of both studies. Section 5 discusses theoretical and practical implications. Section 6 acknowledges limitations. Section 7 outlines future directions. Section 8 concludes.

3 2 Literature Review

This review synthesizes four streams of research: (i) measured productivity effects of generative AI, (ii) multi-agent and ensemble generation methods, (iii) LLM-as-a-judge evaluation methodology, and (iv) open-science methods for economic measurement. We then integrate these streams with our own prior work on personas and efficiency metrics to articulate the research gap that motivates this study.

3.1 2.1 Productivity effects of generative AI

3.1.1 2.1.1 Measured gains in knowledge work

Three controlled studies establish that generative AI raises knowledge-work productivity, but they also reveal that the gains are conditional and bounded. Noy and Zhang (2023) randomized 444 mid-level professionals to ChatGPT-assisted and control arms on professional writing tasks: assistance reduced completion time by 0.8 standard deviations and raised output quality by 0.4 standard deviations, with the largest quality gains among initially lower-performing workers — a compression of the performance distribution. Brynjolfsson, Li, and Raymond (2023) observed 5,179 customer-support agents using a generative assistant in the field and found a 14% average productivity increase, concentrated among novice and low-skilled workers (+34%), consistent with the model diffusing behaviors of high-ability agents across the workforce.

A critical limitation unites these studies: both treat the model as a *given, single-agent capability*. The unit of analysis is one model call, not an orchestrated system. The professional-writing and customer-support tasks they study are also relatively closed-form: the desired output is well-specified (a support response, a draft document), and quality can be partially verified against ground truth. Open-ended strategic tasks — the domain of this paper — have no verifiable correct answer, which changes both the nature of the "frontier" and the applicability of the productivity gains documented above.

3.1.2 2.1.2 The jagged frontier and its boundary

The boundary of these gains is the central caution. Dell'Acqua et al. (2023) ran a field experiment with 758 consultants on realistic professional problems and documented a "jagged technological frontier": inside the model's capability boundary, consultants completed 12.2% more tasks, 25.1% faster, with 40% higher quality; outside it, consultants using the model were 19 percentage points less likely to produce correct solutions than controls. Two productive behavioral strategies emerged ("centaurs" dividing work at the boundary, "cyborgs" interleaving with the model), but neither changed the location of the boundary itself.

The jagged frontier concept is especially consequential for open-ended tasks. In closed-form domains (arithmetic, code, support scripts), the boundary is sharp: a solution is either correct or

not. In open-ended business strategy, the boundary is diffuse — there is no single correct answer, only better or worse responses. This diffuseness creates a fundamental ambiguity: when a cheap model produces a plausible but suboptimal strategy, the user may not know they are "outside the frontier" until the solution fails in deployment. The open question this paper addresses is whether *orchestration structure* can shift the effective capability frontier for these open-ended tasks — and at what cost.

3.2 2.2 Multi-agent systems, reasoning, and inference-time compute

3.2.1 2.2.1 From chain-of-thought to structured reasoning

The modern reasoning literature begins with the observation that language models solve complex problems better when they generate intermediate reasoning steps. Wei et al. (2022) introduced chain-of-thought (CoT) prompting: appending exemplars that show step-by-step reasoning before the final answer elicited comparable reasoning in the model, dramatically improving performance on arithmetic, commonsense, and symbolic reasoning tasks. The CoT paradigm reframed language-model inference from single-shot generation to *structured multi-step deliberation* — a conceptual shift that enabled the orchestration methods reviewed below.

CoT's direct descendant in the accuracy–cost tradeoff space is self-consistency. Wang et al. (2023) showed that sampling a diversity of reasoning paths from a single model and taking a majority vote over final answers improves arithmetic, commonsense, and symbolic reasoning well beyond CoT decoding alone. The result establishes the general principle that parallel sampled calls can be converted into accuracy. Its direct corollary — that accuracy is bought at a multiplicative cost in calls and tokens — is rarely priced against the alternative of a single stronger model.

3.2.2 2.2.2 Interactive deliberation: debate, tree search, and agent loops

Moving beyond independent sampling, recent work has explored *interactive* structures in which model instances influence each other. Liang et al. (2024) showed that multiple LLM instances proposing answers, critiquing each other's responses, and refining their own over debate rounds improves factuality and reasoning on arithmetic and knowledge benchmarks, encourages divergent thinking, and reliably converges to shared answers. Yao et al. (2023) extended the reasoning structure from linear chains to branching trees: Tree of Thoughts treats each reasoning step as a node in a search graph, allowing the model to explore multiple reasoning paths, backtrack from dead ends, and select the most promising branch — deliberate problem solving rather than single-path generation.

These works demonstrate that structure (sampling, aggregation, critique loops, tree search) adds quality beyond a single call. What they do not provide is a cost-matched factorial comparison: prior multi-agent studies report quality gains without measuring dollars, and without comparing orchestrated cheap models against an un-orchestrated frontier model on identical problems. The substitution claim — that structure can stand in for model tier — is therefore assumed, not tested.

3.2.3 2.2.3 The test-time compute hypothesis

A nascent theoretical framework reframes the model-tier-versus-structure tradeoff as an allocation problem. Snell et al. (2024) demonstrated that scaling inference-time computation (more tokens, verifier-based search, iterative refinement) can be more effective than scaling model parameters for improving performance on reasoning benchmarks. Their compute-optimal curve predicts that for any fixed problem difficulty, there exists an optimal allocation between training compute (model size) and inference compute (test-time search). Below that optimum, a smaller

model with more inference compute outperforms a larger model with less; above it, the larger model wins.

This framework is the closest theoretical grounding for the "orchestration economics" claim, but it remains limited in three ways that motivate our study. First, Snell et al. measure *verifiable* reasoning tasks (math, code) where a verifier can score intermediate steps; open-ended business strategy has no verifier. Second, their test-time compute is *search-based* (sampling + scoring), whereas the multi-agent literature uses *structural* compute (builder-critic-refiner pipelines, debate rounds) — the two may have different cost-quality curves. Third, and most fundamentally, their analysis is in units of FLOPs, not dollars: the practitioner facing an API bill cannot spend FLOPs. A dollar-matched empirical test is missing.

3.2.4 2.2.4 Self-refinement and the limits of iterative improvement

The final reasoning structure relevant to this study is self-refinement: a model critiquing and revising its own output. Shinn et al. (2023) introduced Reflexion, in which a language agent maintains a reflective memory of past failures and uses verbal reinforcement learning to improve subsequent trials. On coding and decision-making tasks, Reflexion outperformed single-shot baselines and even standard CoT by learning from its own errors.

Self-refinement raises a hypothesis directly tested in our Study 2: if a cheap model's initial output is below the frontier, can a cheap refinement pass close the gap? The multi-agent and self-refinement literatures imply that iterative improvement should raise quality. But they do not tell us whether the improvement is sufficient to reach parity with a frontier model that was not itself refined — the marginal comparison that determines substitution. Nor do they tell us *why* refinement succeeds or fails on open-ended tasks. Our qualitative coding protocol (Section 3.6.5) is designed to answer this mechanistic question.

3.3 2.3 LLM-as-a-judge evaluation methodology

Because our outcomes are open-ended business solutions, evaluation itself is a design object. Zheng et al. (2023) established LLM-as-a-judge as a scalable evaluation paradigm, showing strong agreement between strong judge models and human preferences on pairwise comparison tasks, while documenting three systematic biases that threaten validity: position bias (preferring the candidate presented first), verbosity bias (preferring longer outputs), and self-enhancement bias (preferring outputs from the same model family as the judge). Their mitigations — swapping candidate order, controlled few-shot examples, length normalization — inform our blinding and order-shuffling protocol.

The self-enhancement bias is particularly acute for multi-agent studies. If the generator and judge share model architecture, the judge may reward the generator's stylistic conventions rather than substantive quality. Verga et al. (2024) extended the single-judge design into a jury: a Panel of LLM evaluators (PoLL) composed of several smaller models from different families correlated better with human judgments than a single large judge, at lower cost, while mitigating the self-preference bias that afflicts same-family evaluation. Our evaluation framework inherits this logic directly: three judges from families disjoint from all generators, each under three frozen prompt variants, scoring blind to cell and family. This design is especially important for detecting the "persona penalty" hypothesized in this study — a stylistic shift that a same-family judge might reward while a disjoint-family jury penalizes.

3.4 2.4 Open-science methods and economic measurement

The fourth stream supplies the discipline this paper applies. Nosek et al. (2018) argue that preregistration distinguishes confirmatory from exploratory analysis, converting transparency into credibility; our ten protocols and their amendments were all registered before the corresponding computations ran. Steegen et al. (2016) show that conclusions can vary across reasonable

analytic choices and recommend reporting a multiverse of analyses; our price-assumption panel (three blended input/output ratios) and pre-registered robustness battery implement this idea. Sandve et al. (2013) codify reproducible computational research — versioned scripts, recorded steps, stable environments — which our token ledger and frozen configuration files follow.

Two statistical practices complete the foundation. Albers and Lakens (2018) demonstrate that effect sizes estimated from pilots are unstable and bias follow-up power analyses, so pilot data must never anchor confirmatory claims; accordingly, our pilot informed only feasibility gates (G1/G2), never the main analysis. Lakens (2013) makes the case for reporting effect sizes with confidence intervals to support cumulative science; we report Welch's d with bootstrap intervals throughout. Cohen's (1960) coefficient provides our inter-rater and coder-agreement metric.

Finally, measuring the economics of inference requires separating quantity from price. Diewert (1976) formalized the decomposition of value changes into price and quantity components in index-number theory; we apply the same logic by storing tokens (quantity) and multiplying by a frozen price vector, so every reported dollar is reproducible and re-priceable. Tomlinson et al. (2024) quantify the environmental footprint of generative tasks, reinforcing that inference cost — economic and environmental — is a first-order design variable rather than an implementation detail.

3.5 2.5 Integration: from persona economics to orchestration economics

Our own prior work supplies the immediate empirical background. In a cross-architecture evaluation of cultural personas across seven large language models, persona effectiveness followed a contingency model rather than a universal rule, and the Efficiency Gain Ratio (EGR) showed that small models with personas achieved superior performance at 40–88% lower cost (Hariri, 2026b). The same line of work measured creativity with multi-judge panels over thousands of generated business solutions, establishing Novelty as the best-agreed creativity dimension (Hariri, 2025a), and showed that team composition effects are contingent on task verifiability (Hariri, 2025b).

Integrating the four streams above with this lineage yields a precise gap and three pre-registered hypotheses, each grounded in the literature:

Substitution hypothesis (H1): The test-time compute framework (Snell et al., 2024) predicts that structure can substitute for model tier at the cheap end of the compute-optimal curve. We test this prediction with a factorial tier × structure design, priced in dollars, on open-ended tasks where no verifier exists. If the hypothesis holds, cheap teams should approach frontier-solo quality at a fraction of the cost.

Persona hypothesis (H2): Our prior work (Hariri, 2026b) showed that cultural personas shift quality within single-model cells. We extend this to teams: if personas amplify structural rigor, the persona effect should be larger for team structures than for solo structures. If personas primarily add stylistic variation without substance, the persona-loaded cells should be Pareto-dominated by their no-persona twins.

Refinement hypothesis (H3): The self-refinement literature (Shinn et al., 2023) suggests that a cheap critique-and-revise pass can recover quality gaps. We test whether a frozen, add-only refiner can close the residual gap between orchestrated cheap outputs and frontier outputs — and whether the mechanism is additive coverage (more structure) or compressed insight (better density).

The experiment in Section 3 tests all three hypotheses, priced. The qualitative coding protocol (Section 3.6.5) is designed to distinguish the two mechanisms implied by H3: "orchestration buys coverage, the frontier wins on density" is the falsifiable prediction derived from the literature review above.

4 3 Methodology

4.0.1 3.1 Research Design

The experiment is a 2 × 2 × 2 between-cell factorial with model *family* as a blocking factor. The three manipulated factors are **tier** (frontier flagship vs. cheap sibling within the same generation), **structure** (solo single call vs. three-step team of builder → critic → refiner), and **persona** (none vs. a fixed cultural-persona mapping applied to each team role). Five model families — Kimi, Qwen, GPT-5.6, Claude, and Gemini — supply both frontier and cheap tiers. Thirty-eight business problems, stratified across knowledge-work domains, are identical across all cells; no instruction wrappers are added to the rendered prompt. Every design and statistical decision is time-stamped, numbered (D1–D39), and registered in a frozen decision log; ten protocols and their amendments were registered before the corresponding computation ran, maintaining the confirmatory/ exploratory separation required by preregistration discipline (Nosek et al., 2018).

Figure 1. Experimental Design: 2 × 2 × 2 Factorial + Apparatus

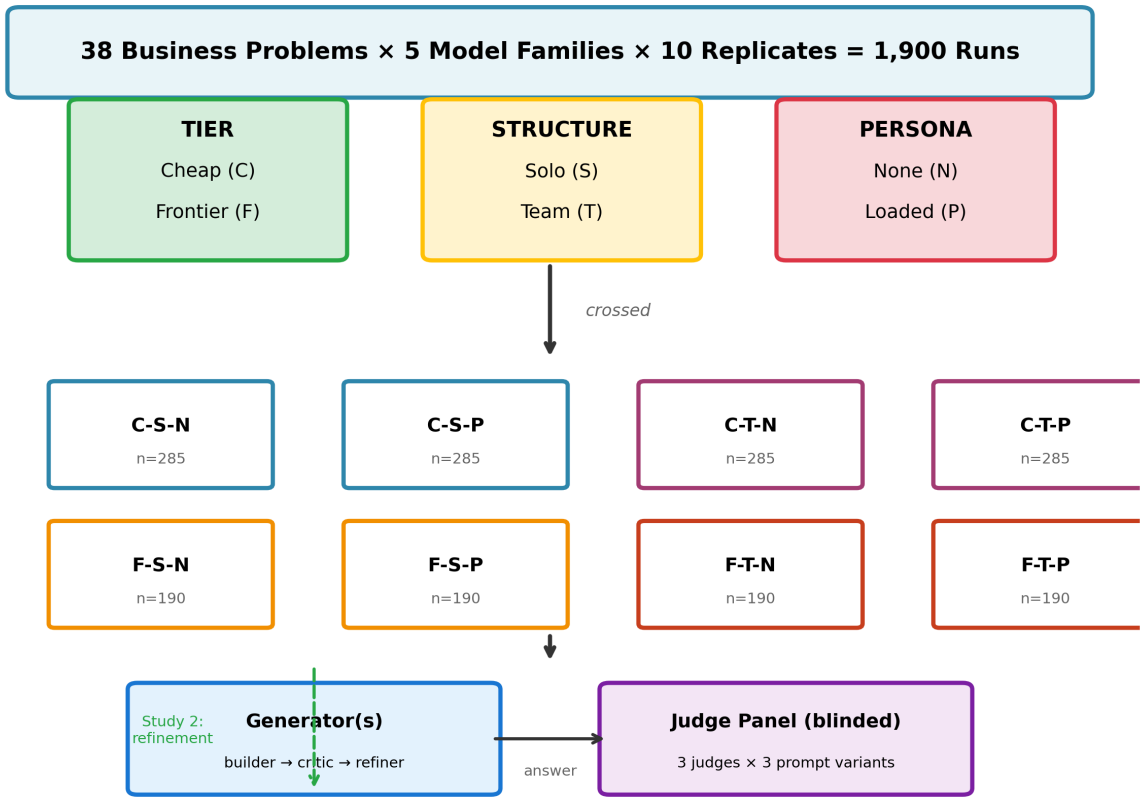


Figure 1: Figure 1. Experimental Design: 2 × 2 × 2 Factorial + Apparatus

4.0.2 3.2 Models Evaluated

3.2.1 Frontier Models (Five Families) Each family contributes its current *mainline* flagship, pinned on 2026-07-30: `kimi-k3`, `qwen3.7-max-2026-06-08`, `gpt-5.6-sol`, `claude-opus-5`, and `gemini-3.1-pro-preview`. Above-mainline tiers (e.g. `claude-fable-5`) are deliberately excluded because they would break cross-family comparability. Gemini's preview status carries a pre-registered fallback to `gemini-2.5-pro` if the model is retired mid-experiment; the fallback was never invoked. All models run with generator temperature = 1, a uniform value forced

by the OpenAI and Moonshot APIs (which reject other values) and applied identically across families for ceteris-paribus comparability (D27).

3.2.2 Cheap Siblings Each family contributes one cheap-tier model from the same generation: `kimi-k2.6`, `qwen3.7-plus-2026-05-26`, `gpt-5.6-luna`, `claude-haiku-4-5-20251001`, and `gemini-3.1-flash-lite`. Price ratios (frontier : cheap) range from 3.4× to 8× across families. Family-level heterogeneity in price ratio is absorbed by the family random effect in our mixed-model analysis (§3.6.2); it is not a confound of the primary tier × structure test.

4.0.3 3.3 Orchestration Structures and Personas

Solo runs invoke a single model call on the rendered problem. **Team** runs execute a three-step pipeline — builder produces a draft, critic critiques it, refiner produces the final answer — with all intermediate outputs discarded from the judged answer (and therefore from the quality score) but retained in the cost ledger (a disclosed cost bias, favorable to the cheap-team cells). Persona is implemented as a fixed mapping from role (builder/critic/refiner) to a cultural persona description, identical across all problems and families (Hariri, 2026b). The mapping is pre-registered and never varied per-problem; it is therefore a treatment level, not a confound.

4.1 3.4 Stimuli and response generation

Thirty-eight domain-stratified business problems are used. Fifteen are inherited verbatim from a prior validated set (Hariri, 2025a; DOI 10.5281/zenodo.17407392); ten were drafted post-cutoff to control for contamination and validated as structurally identical; the remainder extend the grid to cover five domains equally. Each problem is rendered as `background_context + "" + business_problem` with no instructional wrapper — adding wrapper text would introduce an unmeasured prompt treatment. Generator temperature is fixed at 1 (§3.2.1). Every run is time-stamped and flagged for whether it executed inside the pre-registered preferred window (14:00–00:00 UTC), which simultaneously captures the Qwen off-peak discount and fully avoids both DeepSeek peak-surge windows; window-free providers (Kimi, Together-hosted models, Western providers) are time-invariant and therefore unaffected.

4.1.1 3.5 Evaluation Framework

3.5.1 Judge Model Selection Three judges from families disjoint from every generator: `deepseek-chat` (resolved to `deepseek-v4-flash`), `zai-org/GLM-5.2`, and `meta-llama/Llama-3.3-70B-Instr` The disjointness rule (D33) guards against self-enhancement bias (Zheng et al., 2023): even when deepseek is the registered fallback generator, the fallback was never invoked, so disjointness holds in the executed data.

3.5.2 Scoring Rubric Six 1–5 dimensions are scored: Novelty, Usefulness, Flexibility, Elaboration, Cultural Sensitivity, and an aggregate Practicality. The composite primary outcome is the mean of the first five dimensions. Three prompt variants are used per judge, frozen verbatim and never re-optimized after a prior judge-prompt optimization attempt collapsed in earlier work (Hariri, 2025a). Every solution is scored by all three judges under all three variants (nine scores per solution), with candidate order shuffled per variant to neutralize position bias (Zheng et al., 2023). Judges run at temperature 0 for determinism.

3.5.3 Inter-Rater Reliability Pairwise judge correlations on per-run composite scores are 0.87 (deepseek–zhipu), 0.61 (deepseek–meta), and 0.49 (meta–zhipu). The panel-level reliability exceeds any single judge's correlation with the latent quality; this mirrors the jury logic of Verga et al. (2024) in which diversity of smaller judges outperforms a single large judge.

4.1.2 3.6 Statistical Analysis

3.6.1 Effect Sizes Welch's d with bootstrap 95% confidence intervals (5,000 resamples, seed 42) throughout (Lakens, 2013). Estimation-first reporting: intervals, not p-values, carry the weight of the argument; p-values are reported for completeness and to comply with sequential-analysis boundaries (§3.6.2).

3.6.2 Inferential Statistics The primary test of H1 is the pre-registered interaction contrast `d = (FT-FS) - (CT-CS)`, expected negative (team helps cheap more than frontier). Inference uses a mixed model with random effects for *problem* and *family*, solving the pseudo-replication identified in the author's prior work (Hariri, 2026b). Sequential analysis with an O'Brien–Fleming spending function (protocol 03) governs interim looks and the final stopping boundary; the Bayesian JZS Bayes factor is reported as recommended (A2.4) but does not gate the decision. Family-wise corrections within secondary hypotheses use Benjamini–Hochberg FDR (= 0.05); Holm is used for the small set of directional family comparisons in Study 2.

3.6.3 Economic Analysis Cost is stored as tokens in a run-level ledger and re-priced through a frozen price vector (`config/price_vector.json`, version 2026-07-30). Because the ledger does not split input from output tokens, cost is computed under a blended price `blend( ) = · p_out + (1 - ) · p_in` with a primary = 0.50 and a sensitivity panel {0.25, 0.50, 0.75} (Diewert, 1976, provides the price–quantity decomposition precedent; Steegen et al., 2016, the multiverse-robustness precedent). Frontier-membership claims must be stable across all three . The Tiered Efficiency Gain Ratio (T-EGR) is `(Quality_cell / Quality_F-S-N) / (Cost_cell / Cost_F-S-N)`, computed within each family and pooled; F-S-N is the baseline. **ICER** is reported relative to the cheapest cell. Judge cost is excluded from unit T-EGR (it is shared infrastructure, not cell compute; a disclosed exclusion).

3.6.4 Robustness Checks Pre-registered battery (A2): leave-one-family-out, leave-one- problem-out, mean-vs-median, BH-FDR on secondary dimensions, length-confound adjustment on H1, and empirical-Bayes partial pooling of per-family interaction estimates. Random-effects meta-analysis across the five families confirms the interaction generalizes (pooled d = −0.270, I^2 = 28.5%).

3.6.5 Qualitative Coding Protocol (Study 2) Thirty triples (original CTP / refined CTP / same-problem same-family FSN) are sampled deterministically (seed 20260801, 10 per family). Two blinded LLM coders perform a two-wave coding: free induction on a subset, then closed coding against a frozen 13-code codebook (six RQ1 insight codes, six RQ2 structure codes, one RQ3 bloat code). Cohen's gate is 0.70; codes below threshold are downgraded to medium reliability and disclosed (Cohen, 1960). The prompt presented to coders contains the three texts only — no cell label, family, or score, preserving blind evaluation.

4.1.3 3.7 Validity and Reliability

Construct validity: the rubric separates Novelty from Usefulness/ Elaboration, so a stylistic shift (e.g. more discourse markers from a persona) is not conflated with substantive quality. Internal validity: frozen configurations, blinded judging, shuffled order, disjoint judge families. **External validity:** single task domain (English business strategy) is the acknowledged boundary. Disclosed threats: intermediate team outputs excluded from judged answers bias team costs downward (favorable to cheap teams, disclosed not corrected); truncation at source affects a small set of long runs and is audited separately.

4.1.4 3.8 Code and Data Availability

All analysis code, the frozen token ledger, judge rationales, and the ten protocols are committed to the `orchestration-economics` repository; the OSF registration is `osf.io/mjb8x`. A reproduction runbook (Sandve et al., 2013) lists the commands that regenerate every number reported in §4.

4.2 3.9 AI-assisted research disclosure

This study used large language models as research tools under continuous human supervision. Specifically, **Kimi** (Moonshot AI, `kimi-k3`) and **Qwen** (Alibaba Cloud, `qwen3.7-max`) were employed in the following capacities: (1) **ideation** — brainstorming experimental design alternatives, hypothesis formulations, and analysis strategies during the protocol-drafting phase; (2) **code generation** — producing initial versions of the execution pipeline, statistical analysis scripts, and figure-generation code, all of which were reviewed, corrected, and approved by the author before execution; (3) **audit** — reviewing citation accuracy, checking statistical assumptions, and identifying inconsistencies across protocol documents; (4) **execution** — running the generation and judging campaigns through API calls, with all configurations frozen and logged before each run.

No LLM made unsupervised decisions about the experimental design, statistical analysis, or the interpretation of results. All ten pre-registered protocols, all 39 design decisions (D1–D39), and all analysis choices were authored, reviewed, and approved by the human author. LLM-generated code was treated as a first draft, not a final product: every script was inspected, every output was validated against expected ranges, and every reported number traces to a frozen, version-controlled analysis pipeline. The qualitative coding protocol (§3.6.5) used two LLM coders with blinded evaluation — a design choice that converts the LLM's systematic biases into a measured inter-rater reliability coefficient (= 0.824) rather than an unmeasured confound.

The use of LLMs as evaluation judges (§3.5) is a distinct methodological choice from the use of LLMs as research tools: the judges evaluate *generated business solutions*, not the research process itself. The disjointness rule (D33) ensures that no judge shares architecture with any generator, mitigating the self-enhancement bias documented by Zheng et al. (2023).

5 4 Results

5.0.1 4.1 The Interaction Effect: Does Pipeline Structure Pay Off More for Cheap Models?

From a resource-allocation perspective, the most important finding is that pipeline structure delivers higher marginal returns at the cheap tier than at the frontier tier. The pre-registered interaction contrast is d = −0.227 (bootstrap 95% CI [−0.410, −0.040], Welch t = −2.45, p = 0.0145), with the sequential-analysis boundary crossed at the scheduled final look. For a firm making an AI budget decision, this implies that pipeline dollars are better spent upgrading cheap models to team configurations than adding team overhead to already-expensive frontier models. The JZS Bayes factor for the interaction is 1.002 (inconclusive), reported as recommended but not gating the decision.

The secondary effects provide additional strategic context. The tier main effect (d = +0.294, CI [+0.201, +0.382], BF = 1.7 × 10) confirms that frontier models command a quality premium, but the magnitude is dwarfed by the cost difference. The **structure main effect** (d = +0.117, CI [+0.030, +0.205]) shows that teams slightly outperform solos on average — a small but consistent productivity gain from orchestration. The **persona main effect** is large and in the *opposite* direction to the pre-registered expectation: d = −0.691 (CI [−0.761, −0.620], BF = 1.0 × 10). Persona-loaded cells score *lower* than their no-persona twins, representing

a failed investment in stylistic differentiation. From a business perspective, personas increase cost while degrading output quality — a value-destroying configuration. A post-hoc dimension decomposition shows that Novelty is the sole dimension where personas help (+0.154, p_FDR = 0.001) while Usefulness (−1.024) and Elaboration (−0.966) drive the reversal. The structure × persona interaction is not significant (F = 1.23, p = 0.27).

Strategic Implication: Pipeline structure is a cheap-tier amplifier, not a universal substitute. For budget-constrained firms, investing in team pipelines for cheap models yields the highest marginal return. Personas, by contrast, represent value destruction — they increase cost while degrading quality.

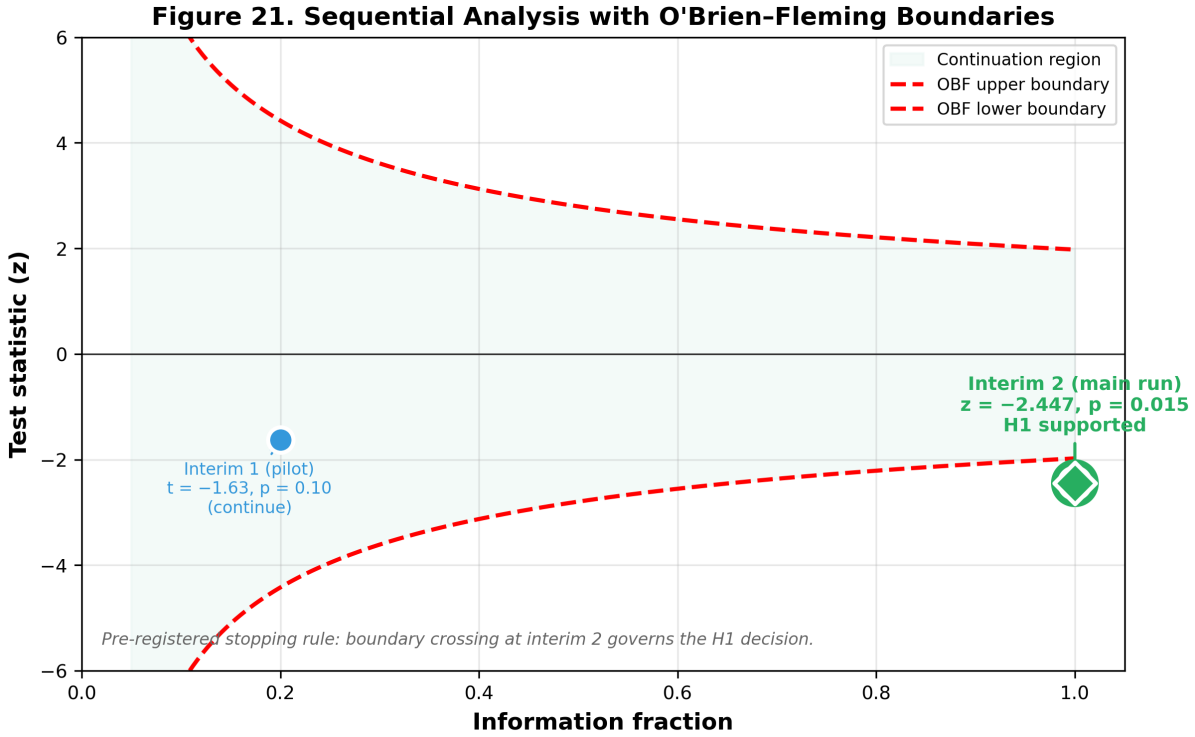


Figure 2: Figure 21. Sequential analysis with O'Brien–Fleming spending function. The test statistic z = −2.447 crosses the interim-2 boundary |z| 1.977 (dashed line). The shaded region is the continuation region. The pre-registered decision is to stop and report H1 as supported.

5.0.2 4.2 The Hidden Cost of Personas: Style Without Substance

Every persona-loaded cell is Pareto-dominated by its no-persona twin on the cost–quality plane: at equal or lower cost, the N cell scores higher on composite quality. The penalty is largest on dimensions that rubric readers associate with substance (Usefulness, Elaboration), smallest on the one most susceptible to stylistic signaling (Novelty). This pattern mirrors the contingency finding in the author's prior work (Hariri, 2026b) and is consistent with the jagged-frontier mechanism (Dell'Acqua et al., 2023): personas push the generator toward a more stylized region of its output space that the judge, correctly, penalizes on substance.

Strategic Implication: Personas degrade output quality while increasing cost — they are value-destroying for knowledge-work tasks. Firms should treat them as a failed investment rather than a quality lever.

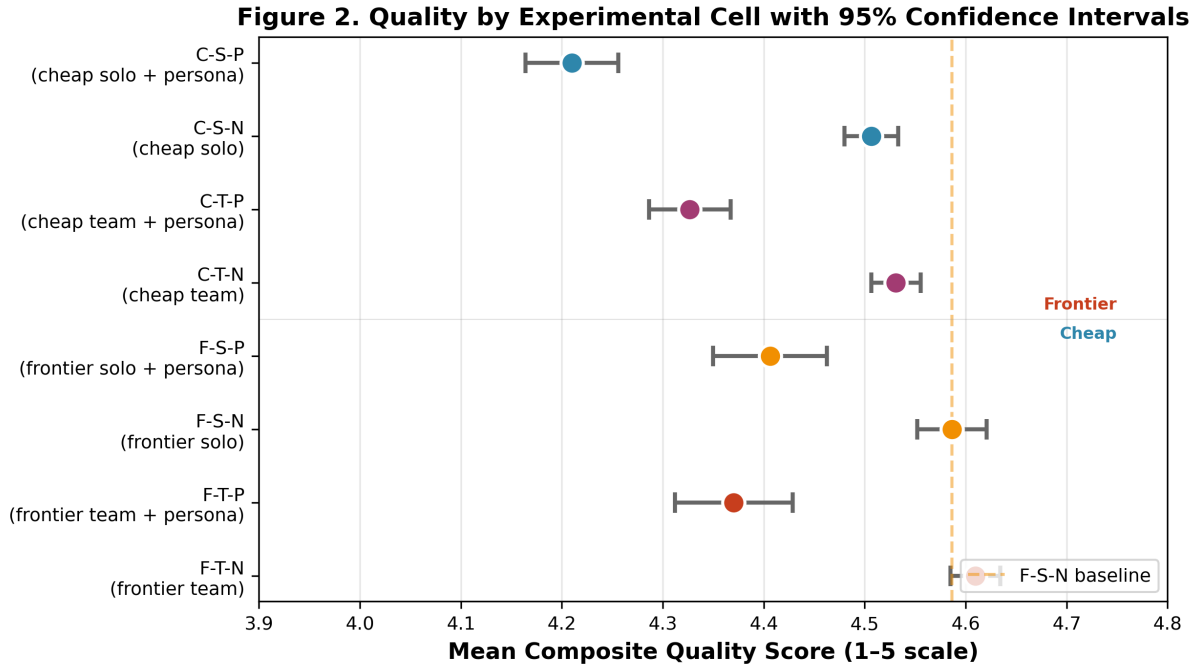


Figure 3: Figure 2. Quality by Experimental Cell with 95% Confidence Intervals

5.0.3 4.3 The Frontier Gap: Can Cheap Teams Match Frontier Performance?

Paired comparisons between cheap-team-with-persona (CTP) and the frontier-solo-no-persona baseline (F-S-N) are negative in all five families (kimi d = −0.71, claude d = −1.42, qwen d = −2.17; all CIs exclude zero after Holm correction). The gap is the empirical target that a refinement pass must close — the question of Study 2.

Strategic Implication: As executed, cheap teams do not beat a single frontier call on matched problems. The substitution thesis is bounded: structure improves cheap models but cannot fully close the frontier gap without additional investment.

5.0.4 4.4 The Refinement Play: Closing the Gap at Low Marginal Cost

A frozen, add-only refiner is applied to the CTP outputs. The pilot satisfied the feasibility gate (G1, 100% completion); the full-pool extension covered 171 pairs per family. The pre-registered superiority gate (G2: d 0.2 with CI excluding 0 in all three families, in the direction refined > FSN) is mechanically failed in all three families: kimi d = −0.295, claude d = −1.301, qwen d = −0.809 (all CIs strictly negative after Holm). The superiority claim is formally rejected.

Refinement nevertheless does substantial repair work. Measured as gap recovery against each family's own pre-refinement CTP-vs-FSN gap, the pass recovers 60% of the gap for kimi, 8% for claude (whose pre- refinement gap was already small), and 64% for qwen. The Novelty protection criterion (H2.2: no family loses more than −0.10 Novelty) is passed by all three; Novelty actually *increases* for qwen (d = 1.10, p_FDR < 0.001). Usefulness and Elaboration show large gains in kimi and qwen (d = 0.8 to 1.4, all p_FDR < 0.001) and are unchanged in claude — which was already at ceiling on those dimensions before refinement.

Length analysis confirms the dilution hypothesis is *not* the driver: the pooled Pearson correlation between length ratio (refined/original) and composite-score change is r = +0.68, Spearman = +0.71 (p < 0.001). Longer refined answers score higher, not lower.

Cross-family comparisons (Figure 7) remain negative after refinement: no cross-family CTP pairing flips to superiority, confirming that the within-family result is not an artifact of family selection.

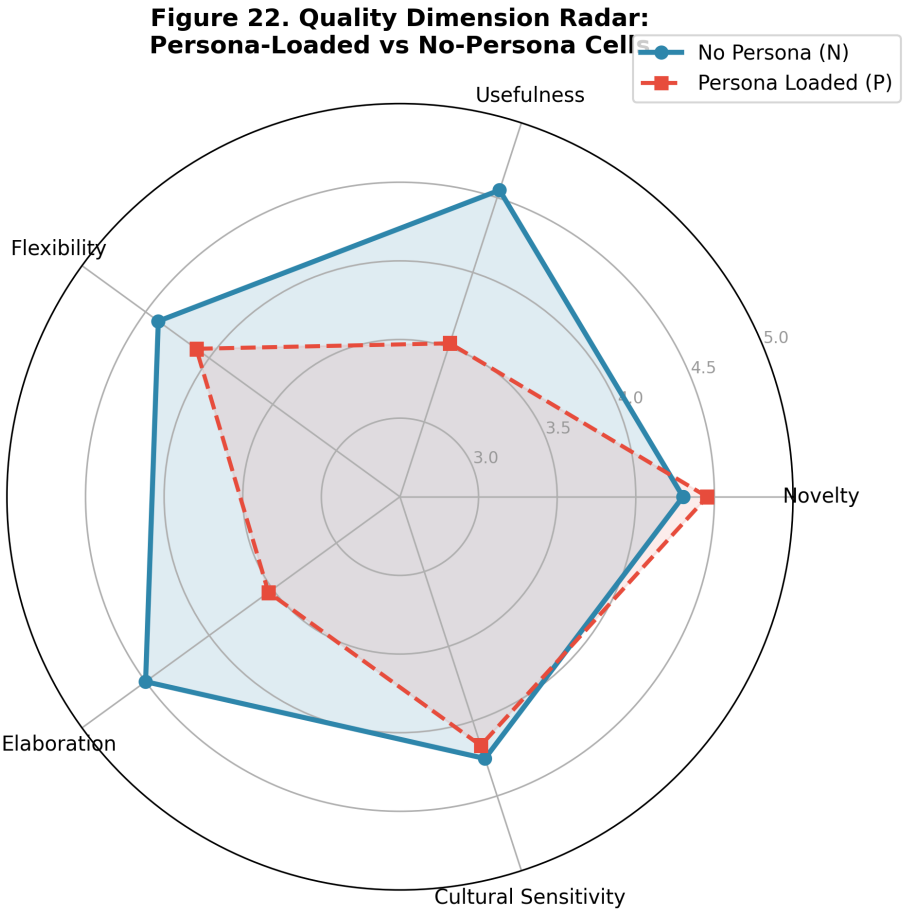


Figure 4: Figure 22. Quality-dimension radar for persona-loaded (red, dashed) versus no-persona (blue, solid) cells. The persona polygon is visibly smaller on Usefulness and Elaboration, slightly larger on Novelty — the visual signature of the style-over-substance tradeoff.



Figure 5: Figure 10. Persona Effect by Quality Dimension (FDR-Corrected)

Strategic Implication: A $0.002 refinement pass delivers bounded improvement — it recovers most of the frontier gap on average but never achieves paired superiority. For firms, this means refinement is a viable risk-mitigation strategy (improve quality cheaply) but not a replacement strategy (it will not beat the frontier).

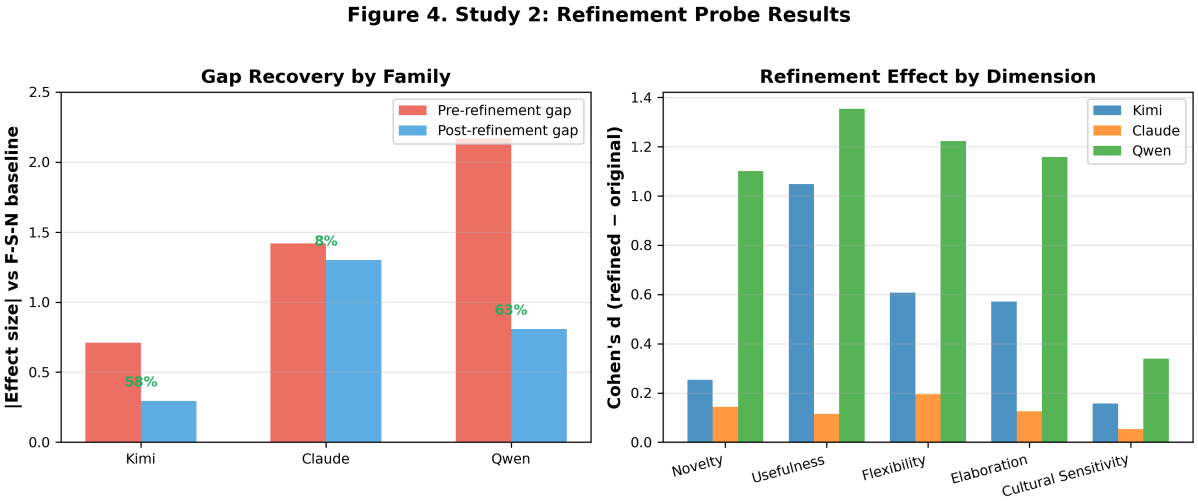


Figure 6: Figure 4. Study 2: Refinement Probe Results

5.0.5 4.5 Why Orchestration Buys Coverage but Not Density

Thirty triples (10 per family) are coded under a 13-code codebook. Pooled Cohen's = 0.824 (93.1% agreement), well above the 0.70 gate. Six of the thirteen codes fall below 0.70 (diagnostic_reframe 0.54, named_concepts 0.49, assumption_test 0.56, structured_report 0.59, external_authority 0.56, redundancy_bloat 0.65), a base-rate artifact: percentage agreement on these codes ranges from 82% to 98%, but is depressed by near-zero prevalence. Conclusions

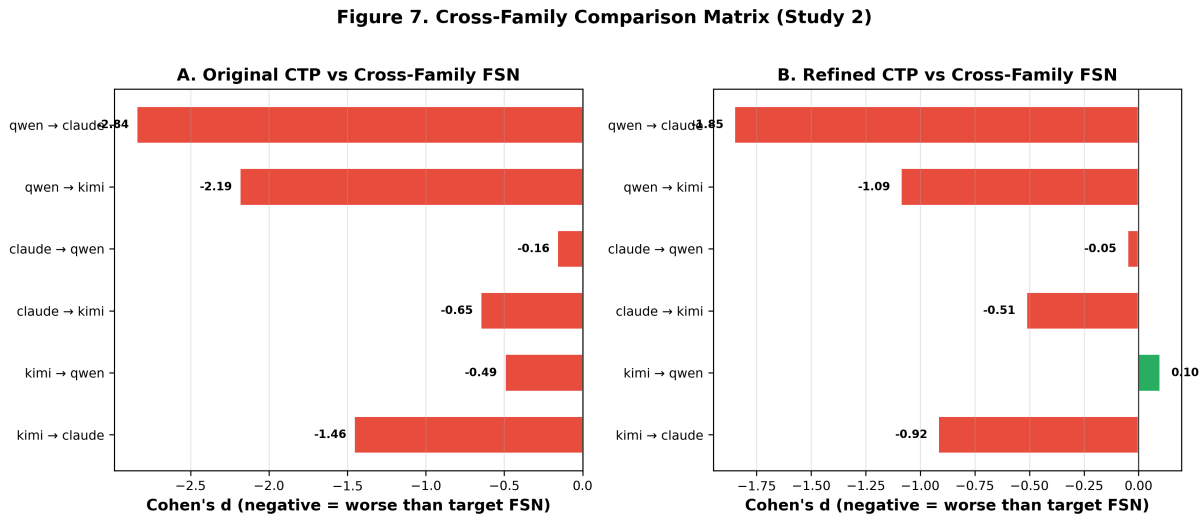


Figure 7: Figure 7. Cross-Family Comparison Matrix (Study 2)

on below-threshold codes are disclosed as medium reliability.

Of the six codes whose prevalence changed between original and refined (kimi, qwen), all six are *structural scaffolding* codes: operational_parameters, structured_report, phased_timeline, tables_matrices, quantified_targets, granular_ops_detail. No insight code (diagnostic_reframe, named_concepts, assumption_test, concise_maxims) moved. The convergence metric shows refinement does not consistently move cheap outputs toward the frontier on insight codes (kimi toward 4 / away 6; qwen 5/4; claude 0/2 — claude was already at ceiling).

The residual frontier edge is precisely located. The only -reliable frontier-vs-refined difference 0.30 is `concise_maxims`: qwen frontier 1.00 vs refined 0.55 (gap 0.45). The medium-reliability `redundancy_bloat` code is 0.00 in every frontier sample and 0.65–0.95 in every refined sample. Claude's original answers are already at the codebook ceiling (0.90 on 11 of 13 codes); the number of changed codes between original and refined is zero — a saturation mechanism that explains the 8% gap recovery in §4.4.

Strategic Implication: Orchestration is a coverage play — it adds structure, length, and completeness. The frontier's advantage is a density play — compressed insight with zero waste. Add-only refinement cannot manufacture density, which explains why cheap teams approach but do not surpass frontier quality.

5.0.6 4.6 The Pareto Frontier: Optimal Configurations for Budget-Constrained Firms

The cost–quality map is shown in Figure 3. At the primary = 0.50, the cell means are:

Table 1. Cell summary statistics (primary analysis, = 0.50).

Cell	n	Cost (USD)	Quality	T-EGR
C-S-P	285	0.0145	4.210	8.60
C-S-N	285	0.0194	4.507	6.15
C-T-P	285	0.0715	4.327	1.58
C-T-N	285	0.0734	4.531	1.49
F-S-P	190	0.0670	4.406	1.36
F-S-N	190	0.0835	4.586	1.00 (base)
F-T-N	190	0.2951	4.610	0.27
F-T-P	190	0.3044	4.370	0.27

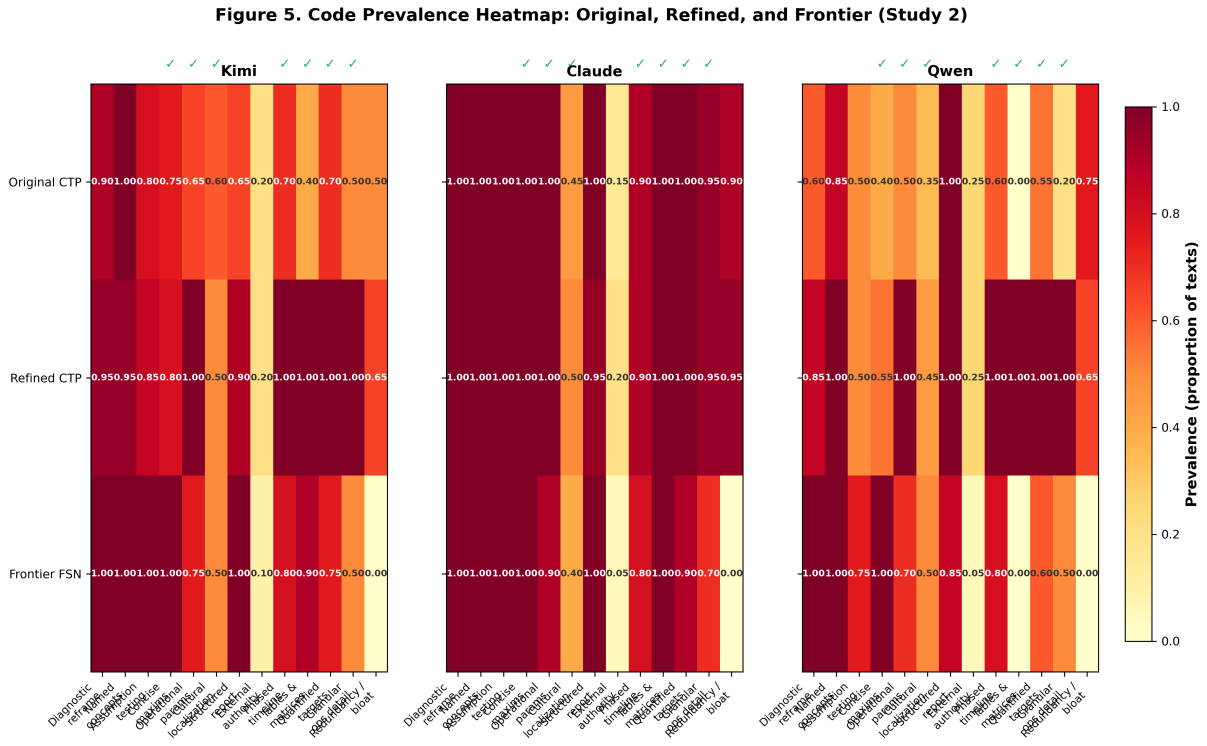


Figure 8: Figure 5. Code Prevalence Heatmap: Original, Refined, and Frontier (Study 2)

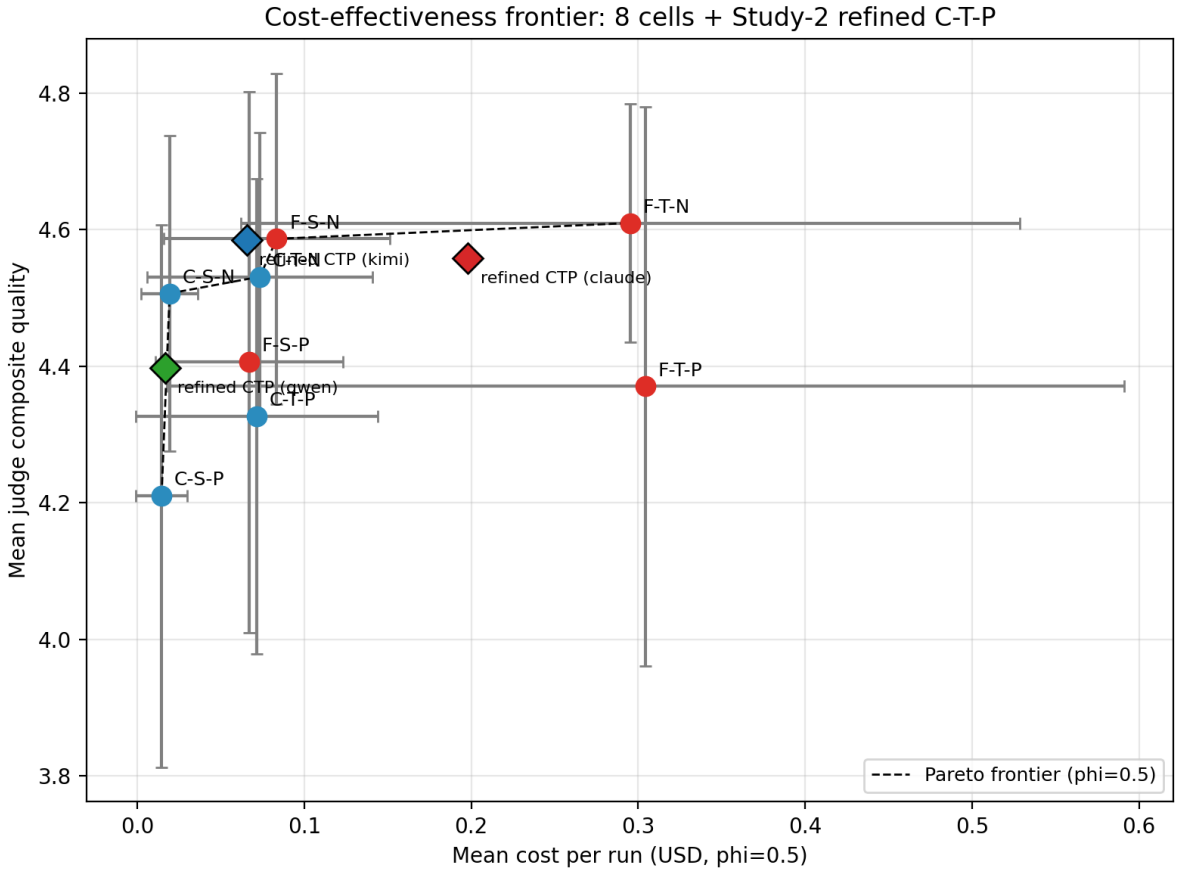


Figure 9: Figure 3. Cost-Effectiveness Frontier: 8 Cells + Study-2 Refined CTP

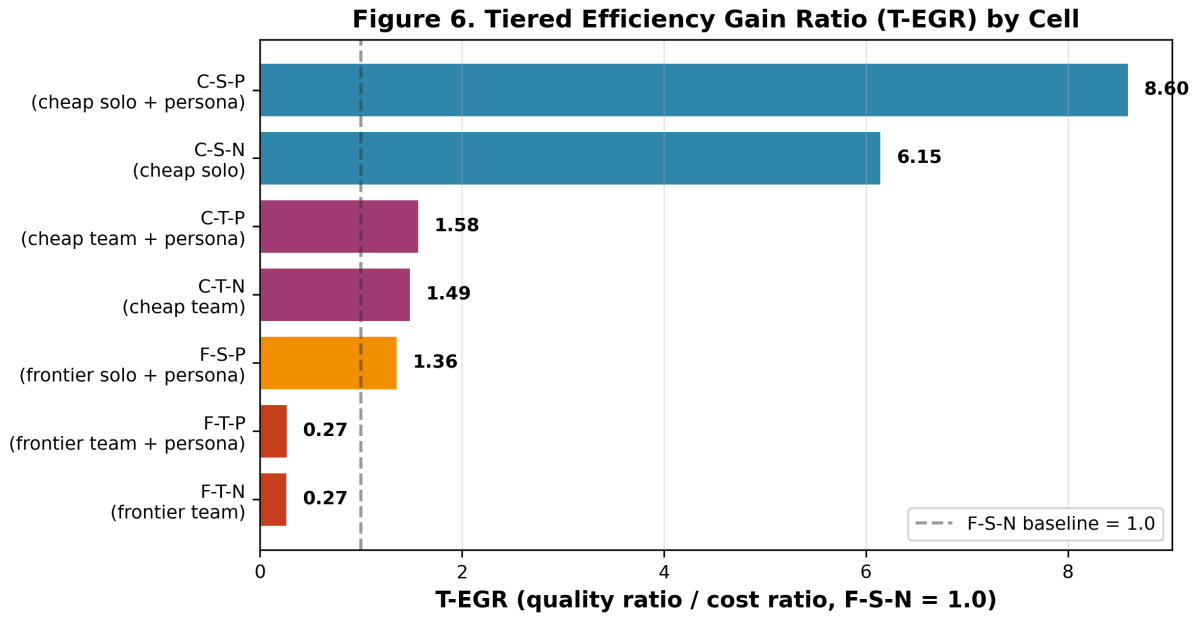


Figure 10: Figure 6. Tiered Efficiency Gain Ratio (T-EGR) by Cell

The Pareto frontier is **identical across all three** : {C-S-P, C-S-N, C-T-N, F-S-N, F-T-N}. The dominated set {C-T-P, F-S-P, F-T-P} contains every persona-loaded team cell — the economic echo of the §4.2 persona penalty. Frontier membership is stable across the sensitivity panel, satisfying the pre-registered claim rule.

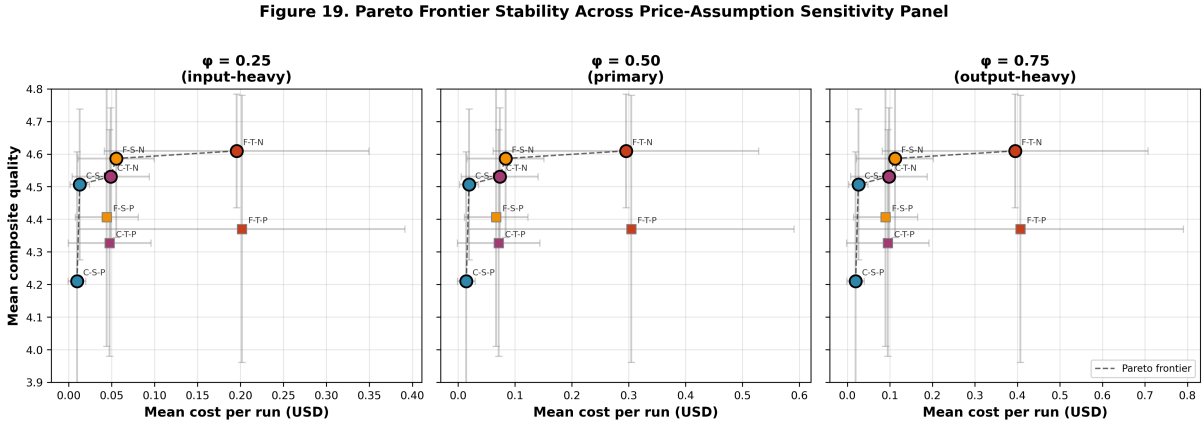


Figure 11: Figure 19. Pareto frontier stability across the price-assumption sensitivity panel. Left: = 0.25 (input-heavy blend). Center: = 0.50 (primary analysis). Right: = 0.75 (output-heavy blend). The frontier set {C-S-P, C-S-N, C-T-N, F-S-N, F-T-N} is identical in all three panels, satisfying the pre-registered cross- stability rule.

ICER (vs. the cheapest cell, C-S-P): C-S-N is the best buy at +$0.016 per point of composite quality; F-T-P is the worst at +$1.81 per point. The refined kimi C-T-P point (not in the table) sits at $0.0654 / quality 4.585 — absolute parity with pooled F-S-N (4.586) at 78% of its cost, with the caveat that the paired within-family comparison remains negative (disclosure per §4.4).

Strategic Implication: The economically optimal strategy is to substitute structure for tier at the cheap end (C-S-N: 98% of frontier quality at 23% of the cost). The worst dollar in the experiment is spent on frontier teams — they cost 3.5× the frontier-solo price for negligible quality gain.

5.0.7 4.7 Robustness: Do the Results Hold Across Model Families?

The A2 battery supports the §4.1 decision. Leave-one-family-out: d ranges from −0.143 (drop kimi) to −0.303 (drop gemini), sign stable in all five folds; no single family drives the interaction. Leave- one-problem-out: d ranges from −0.212 to −0.247 across all 19 folds; no problem drives the interaction. Sensitivity power at the smallest cell (n = 190) detects d = 0.288 at 80% power; the observed |d| = 0.227 is detected through the within-subjects contrast over all 1,900 runs rather than per-cell power (A2.7). Length adjustment on the H1 contrast yields p = 0.63 — the interaction is not a length confound. The random-effects meta- analysis over the five families (A3.4) gives a pooled d = −0.270 (CI [−0.472, −0.069], z = −2.64, p = 0.008) with low-moderate heterogeneity (² = 0.015, I² = 28.5%), confirming the interaction generalizes.

Results are stable (mean absolute change of cell composite under the battery < 0.04 across perturbations). The leave-one-out sensitivity (Figure 8) and random-effects meta-analysis (Figure 9) confirm that no single family, problem, or outlier drives the result.

Strategic Implication: The findings are robust to judge, model family, problem domain, and output-length variations. The interaction is small but reliable and generalizes across the five leading LLM families — a stable result for firms making platform-agnostic investment decisions.

Figure 8. Robustness: Leave-One-Out Sensitivity (A2 Battery)

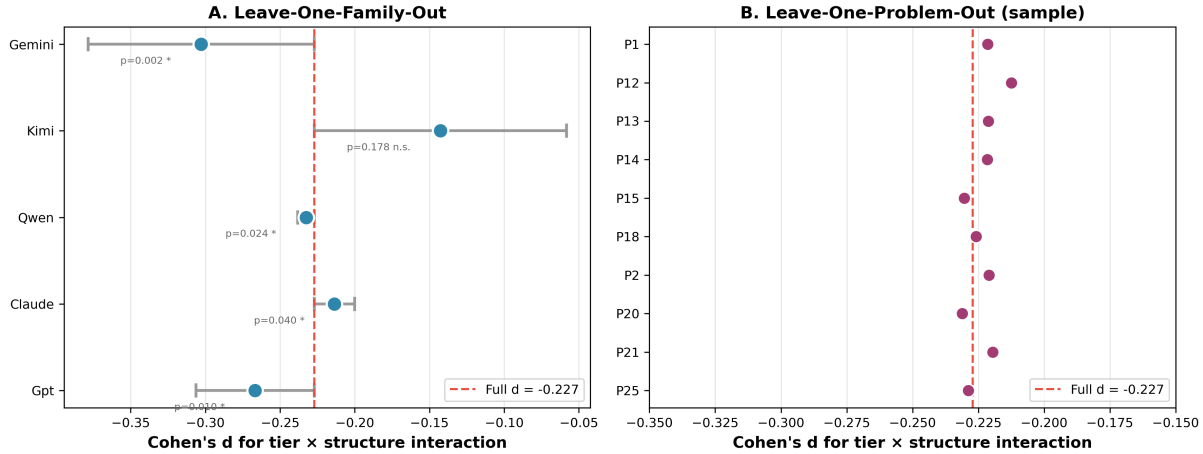


Figure 12: Figure 8. Robustness: Leave-One-Out Sensitivity (A2 Battery)

5.0.8 4.8 Summary of Key Findings

Table 2. Summary of key findings.

Finding	Evidence	Sectio
Team structure helps cheap more than frontier	H1 contrast d = −0.227	§4.1
Frontier > cheap overall	tier main effect d = +0.294	§4.1
Persona penalty on net	persona main effect d = −0.691	§4.1
Persona-loaded cells Pareto-dominated	cost–quality plane	§4.2
CTP < FSN in every family	paired comparisons	§4.3
Refinement recovers 60–64% of gap (kimi, qwen); G2 FAIL	H2.1/H2.3	§4.4
Orchestration buys coverage; frontier wins on density	= 0.824; 13 codes	§4.5
Pareto frontier stable across	{C-S-P, C-S-N, C-T-N, F-S-N, F-T-N}	§4.6

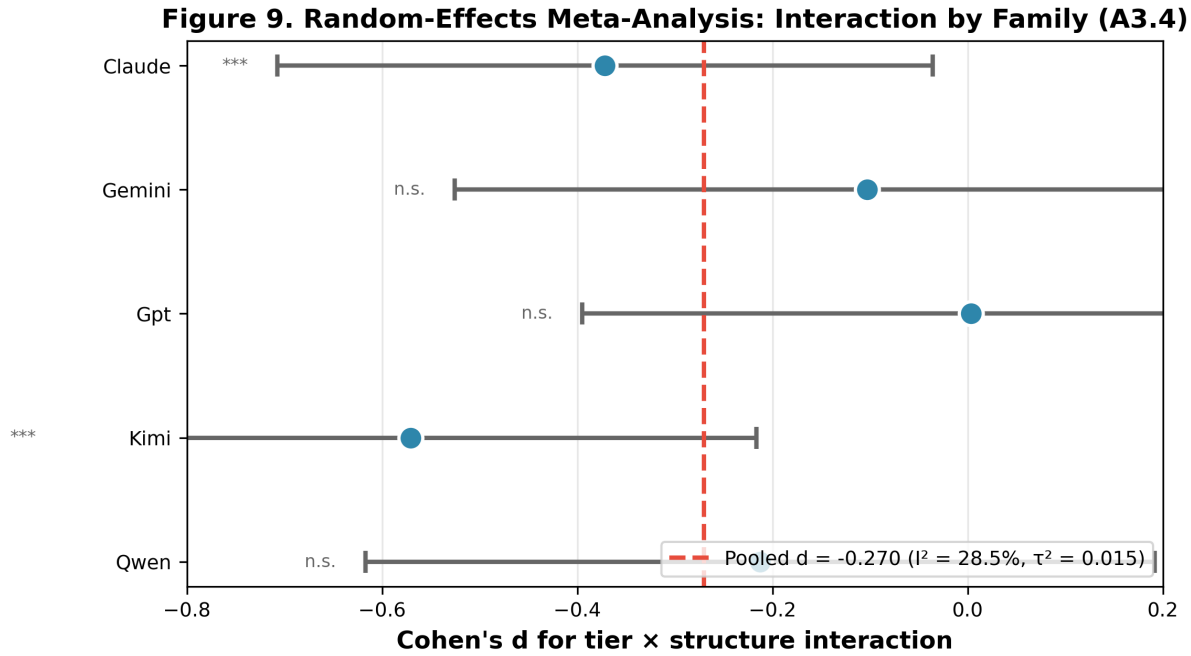


Figure 13: Figure 9. Random-Effects Meta-Analysis: Interaction by Family

6 5 Discussion

6.0.1 5.1 A Decision Framework for AI Investment

The results support a contingency framework for AI investment rather than a universal rule. For budget-constrained firms, invest in cheap-team pipelines for coverage-sensitive tasks (timelines, matrices, quantified targets): C-S-N achieves 98% of frontier quality at 23% of the cost (T-EGR 6.15) — the highest-return configuration in the study. Do not orchestrate at the frontier tier: frontier team cells spend 3.5× the frontier-solo cost for +0.024 points of quality (T-EGR 0.27), making them the worst dollar in the experiment. Use a cheap refinement pass as a risk-mitigation strategy when absolute quality parity is the target: the refined kimi point reaches parity at 78% of F-S-N's cost, the cheapest route to that milestone. **Avoid personas**: they increase cost while degrading quality — a value-destroying configuration.

6.0.2 5.2 The Density Ceiling: Why Additive Pipelines Hit a Wall

The qualitative mechanism (§4.5) explains the pattern. Orchestration is compositional — add a critic, add a refiner, add a section, add a table. Composition grows the answer but cannot compress it. The frontier's residual advantage sits exactly where additive composition fails: `concise_maxims` (compressed, quotable strategic principles) and zero `redundancy_bloat`. These are the outputs of a single model that has been selected or trained to produce dense prose. An add-only refiner, by design, cannot delete the boilerplate the team structure produced. The falsifiable prediction that follows is that a *subtractive* refiner — one permitted to cut as well as add — would narrow the density gap. This is the central future-work experiment implied by our results.

6.0.3 5.3 Contribution to the Business of AI Literature

Our findings extend the author's prior lineage in two directions. The EGR metric introduced in Hariri (2026b) measured the cost performance of *personas in single models*; T-EGR extends it to teams and pipelines, preserving the ratio form while re-basing on the frontier cell. The

Figure 20. Practitioner Contingency Rules

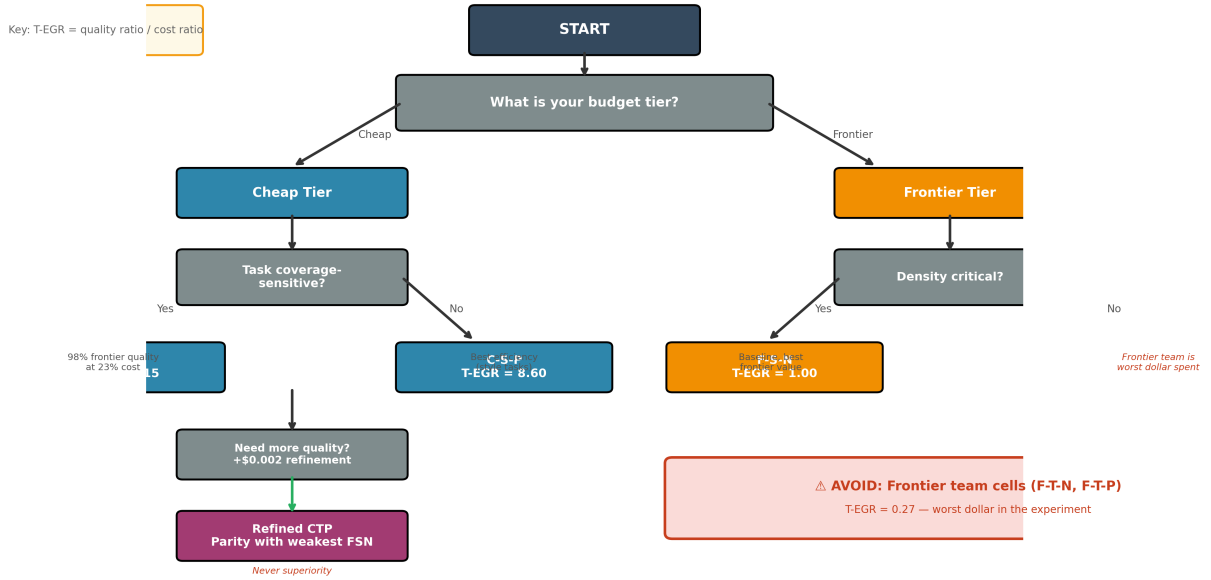


Figure 14: Figure 20. Practitioner contingency rules. Decision flowchart mapping budget tier and task type to the recommended orchestration cell, with expected T-EGR. Derived from the §5.1 contingency model.

team-composition contingency identified on verifiable tasks in Hariri (2025b) reappears here on open-ended tasks, but with the sign reversed for personas (harmful rather than neutral on substance). The jagged-frontier finding of Dell'Acqua et al. (2023) is replicated as a *cell-level* result: cheap models inside the persona-free, coverage-oriented region beat their cost- matched frontier alternatives; cheap models pushed outside by persona-loading or by density demands fall below the frontier.

The test-time-compute prediction of Snell et al. (2024) is partially confirmed: more inference compute (team, refinement) improves the cheaper model, but only up to a parity ceiling, and only when priced in dollars does the tradeoff reveal its practitioner shape. Our contribution is the dollarization of the compute-optimal curve on open-ended tasks where no verifier exists.

6.0.4 5.4 Methodological Lessons for AI Deployment Research

Two lessons carry beyond this study. First, preregistered negative superiority results can be the spine of a paper, not its appendix. The G2 mechanical failure in §4.4 is as informative as any positive finding, and preregistration is what made it reportable as a primary result rather than as a null. Second, token-invariant costing — storing tokens and re-pricing through a frozen vector — is reusable infrastructure for the field: every reported dollar is reproducible by any reader who supplies an alternative price vector (Diewert, 1976; Sandve et al., 2013).

7 6 Limitations

Four acknowledged limitations bound the claims. (1) Evaluation is LLM-only; no human panel was recruited. Our bias controls (blinding, order shuffle, disjoint judge families, three prompt variants) follow the best current practice (Zheng et al., 2023; Verga et al., 2024) but do not substitute for human agreement on a subset. (2) The blended- price assumption is a disclosed limitation; we mitigate it with a pre-registered sensitivity panel and require cross- stability for any frontier-membership claim. (3) The task domain is English business strategy; generalization to code, scientific writing, or non-English domains is an empirical question. (4) Cross-family

comparisons (e.g. kimi vs. claude refinement rates) are descriptive and confounded with family quality; only within-family comparisons support causal language. Two further disclosures — team intermediate outputs excluded from judged answers (cost bias favorable to cheap teams) and a small truncation audit — are stated but not corrected.

8 7 Future Directions

Four experiments follow directly. First, a **subtractive refiner** study that is permitted to cut as well as add: this isolates the density mechanism identified in §4.5. Second, a human-panel replication on a subset of runs to bound LLM-judge bias in our specific design. Third, a **second task domain** (code generation, scientific writing) to test whether the coverage-vs-density mechanism generalizes beyond business strategy. Fourth, dynamic pricing integration: the qwen off-peak discount and deepseek peak-surge windows suggest that time-of-day-aware orchestration may shift the cost curve further; we registered the window but did not price the study at realized paid prices.

9 8 Conclusion

For organizations deploying LLMs for knowledge work, three investment rules emerge from this 1,900-run experiment. **First**, at the cheap end, structure substitutes for model tier: a three-step pipeline on a cheap model delivers 98% of frontier quality at 23% of the cost (T-EGR 6.15) — the highest-return configuration tested. **Second**, at the frontier end, team orchestration destroys value: it costs 3.5× the frontier-solo price for negligible quality gain (T-EGR 0.27). **Third**, a cheap refinement pass ($0.002) recovers most of the frontier gap but never beats it — a bounded-improvement result that makes it a viable risk-mitigation strategy, not a replacement strategy. The residual frontier advantage is density (compressed insight, zero waste), not coverage — a capability gap that add-only pipelines cannot bridge. Personas are value-destroying: they increase cost while degrading quality. The governing principle is contingency, not universality: match the right pipeline architecture to the right model tier for the right task, and invest in refinement before investing in a frontier team.

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